if we have to make a set of 6 order pairs or coordinates and we can select the coordinate numbers from 1 to 6 only make sets of (x,y) {} order pairs can be (4,4) or (6,4) or (4,6) etc etc means x can choose 6 numbers and y casn choose 6 numbers tell me how many different sets of order pairs we can create

explai in details step by step

Treat each ordered pair as a coordinate (x, y), where

- x can be any number from 1 to 6
- y can also be any number from 1 to 6

So the possible coordinates are things like:

$$(1,1), (1,2), \ldots, (4,4), (6,4), (4,6), \ldots, (6,6)$$

Now let's count them step by step.

Step 1: Count choices for x

x can be chosen from:

$$1, 2, 3, 4, 5, 6$$

So there are:

$$6 \text{ choices for } x$$

Step 2: Count choices for y

For each choice of x, y can also be:

$$1, 2, 3, 4, 5, 6$$

So there are:

$$6 \text{ choices for } y$$

Step 3: Multiply the choices

Because each x can be paired with each y, total ordered pairs are:

$$6 \times 6 = 36$$

So the number of different ordered pairs is:

$$\boxed{36}$$

Step 4: Why multiplication works

Think of it like this:

- If $x = 1$, possible pairs are
  $(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)$ → 6 pairs
- If $x = 2$, again 6 pairs
- If $x = 3$, again 6 pairs
- ...
- If $x = 6$, again 6 pairs

So total:

$$6 + 6 + 6 + 6 + 6 + 6 = 36$$

Step 5: Full list of all pairs

$$(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)$$
$$(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)$$
$$(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)$$
$$(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)$$
$$(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)$$
$$(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)$$

That is all 36 ordered pairs.

Important note about your wording: “set of 6 ordered pairs”

This phrase can mean two different things:

Meaning 1: “How many different ordered pairs are possible?”

That answer is:

$$\boxed{36}$$

Meaning 2: “How many different sets containing 6 ordered pairs can be made from those 36 possible pairs?”

Then the answer is choosing 6 distinct pairs from 36:

$$\binom{36}{6} = 1,947,792$$

So:

- total possible individual coordinates: $\boxed{36}$
- total possible sets of 6 distinct coordinates: $\boxed{1,947,792}$

If you want, I can also explain the second answer step by step in the same detailed way.

we use combination because order doesnot matter
but if order matters in sets and and in corrdinates then
and if order matters in corrdinate not in sets
and if order matters in sets but not in coordinates

explain in details step by step

Yes — the answer changes depending on **where order matters**.

You have:

- numbers available: $1, 2, 3, 4, 5, 6$
- a coordinate is made from two numbers
- then you want a **set of 6 coordinates**

There are **three layers** where order may or may not matter:

1. **inside a coordinate**
   Is $(2, 5)$ different from $(5, 2)$?
2. **among the 6 coordinates chosen**
   Is

$$\{(1, 1), (2, 2), (3, 3)\}$$

the same as

$$\{(3,3),(1,1),(2,2)\}?$$

3. **repetition allowed or not**

   Can the same coordinate appear twice?

Usually, when someone says **set of 6 coordinates**, that means:

- no order among the 6 chosen items
- no repetition of the same item

So I'll explain all the cases carefully.

Step 1: First count the possible coordinates

Case A: order matters in a coordinate

Then (x,y) and (y,x) are different unless $x=y$.

Since:

- 6 choices for x
- 6 choices for y

total coordinates:

$$6\times 6=36$$

So there are:

$$\boxed{36}$$

possible ordered coordinates.

Case B: order does not matter in a coordinate

Then $(2,5)$ and $(5,2)$ are the same.

So now each coordinate is really an **unordered pair** from $\{1,2,3,4,5,6\}$, with repetition allowed because $(4,4)$ is allowed.

This is the number of ways to choose 2 numbers from 6 **with repetition**:

$$\binom{6+2-1}{2} = \binom{7}{2} = 21$$

So there are:

$$\boxed{21}$$

possible coordinates if order inside the coordinate does not matter.

You can also see it by listing:

- with 1: $(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)$ → 6
- with 2: $(2,2), (2,3), (2,4), (2,5), (2,6)$ → 5
- with 3: 4
- with 4: 3
- with 5: 2
- with 6: 1

Total:

$$6 + 5 + 4 + 3 + 2 + 1 = 21$$

Step 2: Now choose 6 coordinates

Now we build a collection of 6 coordinates.

Case 1: Order matters in coordinates and order does NOT matter in the set

This means:

- $(2,5) \neq (5,2)$
- but the arrangement of the 6 chosen coordinates does not matter

So:

- number of available coordinates = 36
- choose 6 of them

- order of choosing those 6 does not matter

Therefore use combination:

$$\binom{36}{6}$$

Compute:

$$\binom{36}{6} = 1,947,792$$

Answer:

$$\boxed{1,947,792}$$

Why combination here?

Because once you choose the 6 coordinates, rearranging them does not create a new set.

For example:

$$\{(1,2),(3,4),(5,6),(2,2),(4,1),(6,3)\}$$

is the same set as

$$\{(6,3),(4,1),(1,2),(3,4),(2,2),(5,6)\}$$

So order among the 6 chosen items does not matter.

Case 2: Order matters in coordinates and order matters in the set

This means:

- $(2,5) \neq (5,2)$
- and the position of the coordinates in the 6-place list matters

So now you are not choosing a set, but really an **ordered arrangement of 6 distinct coordinates**.

That means use permutation:

$$^{36}P_6 = \frac{36!}{(36-6)!} = \frac{36!}{30!}$$

Compute step by step:

$$36 \times 35 \times 34 \times 33 \times 32 \times 31$$

$$= 1,402,410,240$$

Answer:

$$\boxed{1,402,410,240}$$

Why permutation here?

Because changing the order of the 6 coordinates gives a different result.

For example:

1st list:

$$[(1,2),(3,4),(5,6),(2,2),(4,1),(6,3)]$$

2nd list:

$$[(6,3),(4,1),(1,2),(3,4),(2,2),(5,6)]$$

These are counted as different.

Case 3: Order does NOT matter in coordinates and order does NOT matter in the set

This means:

- $(2,5) = (5,2)$
- rearranging the 6 chosen coordinates does not matter

So:

- available coordinates = 21
- choose 6 of them
- order among the 6 chosen coordinates does not matter

Therefore:

$$\binom{21}{6}$$

Compute:

$$\binom{21}{6} = 54,264$$

Answer:

$$\boxed{54,264}$$

Case 4: Order does NOT matter in coordinates but order matters in the set

This means:

- $(2,5) = (5,2)$
- but the order of the 6 chosen coordinates matters

So:

- 21 possible coordinates
- arrange 6 distinct ones in order

Therefore:

$$^{21}P_6 = \frac{21!}{15!}$$

Step by step:

$$21 \times 20 \times 19 \times 18 \times 17 \times 16$$

$$= 39,070,080$$

Answer:

$$\boxed{39,070,080}$$

Summary table

If repetition of coordinates is NOT allowed

Order in coordinate?	Order in set?	Count
matters	does not matter	$\binom{36}{6} = 1,947,792$
matters	matters	${}^{36}P_6 = 1,402,410,240$
does not matter	does not matter	$\binom{21}{6} = 54,264$
does not matter	matters	${}^{21}P_6 = 39,070,080$

Very important idea

There are two separate decisions:

A. How many different coordinate types exist?

- if coordinate order matters: 36
- if coordinate order does not matter: 21

B. How do we choose 6 of them?

- if order among the 6 does not matter: use **combination**
- if order among the 6 matters: use **permutation**

So the pattern is:

- **ordered coordinates** → start from 36
- **unordered coordinates** → start from 21

then

- **unordered collection of 6** → choose with $\binom{n}{6}$
- **ordered collection of 6** → arrange with ${}^{n}P_6$

One more intuitive example

Take only 3 coordinates to make it easy.

Suppose chosen coordinates are:

$$(1,2),(3,4),(5,6)$$

If set order does not matter

Then all of these are the same:

$$\{(1,2),(3,4),(5,6)\}$$

$$\{(5,6),(1,2),(3,4)\}$$

$$\{(3,4),(5,6),(1,2)\}$$

So use combination.

If set order matters

Then these are all different:

$$[(1,2),(3,4),(5,6)]$$

$$[(5,6),(1,2),(3,4)]$$

$$[(3,4),(5,6),(1,2)]$$

So use permutation.

If coordinate order does not matter

Then:

$$(1,2)=(2,1)$$

So you do not count both separately.

If coordinate order matters

Then:

$$(1,2)\neq(2,1)$$

So both count separately.

Final answers

Assuming you choose **6 distinct coordinates**:

- **Order matters in coordinates, not in set**:

$$\boxed{\binom{36}{6} = 1,947,792}$$

- **Order matters in coordinates, and in set**:

$$\boxed{{}^{36}P_6 = 1,402,410,240}$$

- **Order matters in set, but not in coordinates**:

$$\boxed{{}^{21}P_6 = 39,070,080}$$

- **Order matters neither in set nor in coordinates**:

$$\boxed{\binom{21}{6} = 54,264}$$

If you want, I can also explain the **same four cases when repetition is allowed**, which gives another set of answers.

↪ This is the number of ways to choose 2 numbers from 6 **with repetition**: (6+2−12)\= (72)\=21\\binom{6+2-1}{2}=\\binom{7}{2}=21(26+2−1)\=(27)\=21

why we do this what this means explain in details

You are asking **why we use**

$$\binom{6+2-1}{2} = \binom{7}{2} = 21$$

to count coordinates when **order in the coordinate does not matter** but **repetition is allowed** (like $(4,4)$).

Let's explain **very carefully step-by-step** what this means and **why this formula works**.

---

1. What the problem becomes

Numbers available:

$$1, 2, 3, 4, 5, 6$$

A coordinate normally looks like:

$$(x, y)$$

But you said **order does not matter**, meaning:

$$(2, 5) = (5, 2)$$

So instead of **ordered pairs**, we are really choosing **two numbers from the set {1..6}**.

Example valid coordinates:

$$(1, 1)$$

$$(1, 4)$$

$$(3, 6)$$

But **(2,5) and (5,2) count as the same**.

So the question becomes:

> How many ways can we choose **2 numbers from 6 numbers** if repetition is allowed?

---

2. Why repetition is allowed

Because coordinates like

$$(4, 4)$$

are allowed.

So we must count pairs like:

$$(1, 1), (2, 2), (3, 3)$$

This means **the same number can appear twice**.

3. First understand by listing

Let's list systematically.

Start with **1**:

$$(1,1)$$

$$(1,2)$$

$$(1,3)$$

$$(1,4)$$

$$(1,5)$$

$$(1,6)$$

Total = **6**

Now start with **2**

We cannot write **(2,1)** because it is the same as **(1,2)**.

So we start from:

$$(2,2)$$

$$(2,3)$$

$$(2,4)$$

$$(2,5)$$

$$(2,6)$$

Total = **5**

Now start with **3**

$$(3,3)$$

$$(3,4)$$

$$(3,5)$$

$$(3,6)$$

Total = **4**

Now start with **4**

$$(4,4)$$

$$(4,5)$$

$$(4,6)$$

Total = **3**

Now start with **5**

$$(5,5)$$

$$(5,6)$$

Total = **2**

Now start with **6**

$$(6,6)$$

Total = **1**

---

4. Total number of pairs

Add them:

$$6+5+4+3+2+1$$

$$=21$$

So there are **21 coordinates** if order does not matter.

---

5. Why the formula works

Instead of listing, mathematics uses a formula for **combinations with repetition**.

The formula is

$$\binom{n+r-1}{r}$$

where

- n = number of different objects
- r = number chosen

In our problem

We choose:

- $r = 2$ numbers
- from $n = 6$ numbers

So

$$\binom{6+2-1}{2}$$

$$= \binom{7}{2}$$

---

6. What does $6 + 2 - 1$ mean?

This comes from a method called **stars and bars**.

Imagine we are distributing **2 selections** among **6 numbers**.

We use:

- **2 stars** (selected numbers)

- **5 bars** (dividers between numbers)

Example representation:

```
* | * | | | |
```

This means:

- one 1
- one 2

coordinate → (1,2)

Another example:

```
** | | | | |
```

This means:

two 1's → (1,1)

Another example:

```
| | * | * | |
```

This means:

one 3 and one 4 → (3,4)

Total symbols:

- stars = 2
- bars = 5

Total objects:

$$2 + 5 = 7$$

So we choose **2 positions for stars among 7 positions**.

That is

$$\binom{7}{2}$$

---

7. Compute the value

$$\binom{7}{2} = \frac{7!}{2!5!}$$

$$= \frac{7 \times 6}{2 \times 1}$$

$$= 21$$

---

8. Final meaning

$$\binom{6+2-1}{2}$$

means:

> Number of ways to choose **2 items from 6 types when repetition is allowed and order does not matter**.

In this coordinate problem it represents:

> The number of **distinct coordinates** if
> $(x, y) = (y, x)$.

Result:

$$\boxed{21}$$

If you want, I can also show a **very powerful geometric trick using a 6×6 grid** that makes the answer **21 obvious in 5 seconds** (this is how mathematicians visualize it).

what if we have {(1,2),(3,4),(5,6),(4,3),(2,1),(6,5)} this set and just want to know how many way it we can arrange

If you treat

$$\{(1,2),(3,4),(5,6),(4,3),(2,1),(6,5)\}$$

as **6 separate items**, and you want to know how many ways to arrange these 6 items, then:

Step 1: Check whether all 6 items are different

As **ordered pairs**:

- $(1,2) \neq (2,1)$

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- $(5,6) \neq (6,5)$

So yes, all 6 are different.

Step 2: Arrange 6 distinct items

Number of arrangements of 6 distinct objects is:

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$= 720$$

Answer

$$\boxed{720}$$

Why?

Because for arrangement:

- 1st place: 6 choices
- 2nd place: 5 choices
- 3rd place: 4 choices

- 4th place: 3 choices
- 5th place: 2 choices
- 6th place: 1 choice

So:

$$6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

Important note

This answer is correct **if order matters inside coordinates**.

That means:

- $(1,2)$ and $(2,1)$ are counted as different
- $(3,4)$ and $(4,3)$ are counted as different
- $(5,6)$ and $(6,5)$ are counted as different

But if order does NOT matter inside coordinates

Then:

- $(1,2) = (2,1)$
- $(3,4) = (4,3)$
- $(5,6) = (6,5)$

So your collection really becomes just:

$$\{(1,2), (3,4), (5,6)\}$$

Now there are only **3 distinct items**, so the number of arrangements is:

$$3! = 6$$

So in that case the answer would be:

$$\boxed{6}$$

Final summary

- **If coordinate order matters**: $\boxed{720}$
- **If coordinate order does not matter**: $\boxed{6}$

In your written set, since you included both $(1, 2)$ and $(2, 1)$, the usual interpretation is that you are treating them as different, so the intended answer is:

$$\boxed{720}$$

I can also show all 720 counting logic using permutation notation step by step.